

光伏瓦屋面系统安装指导书

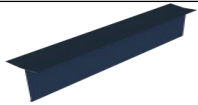
SOLAR ROOF TILE SYSTEM INSTALLATION INSTRUCTIONS

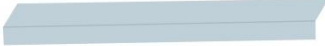
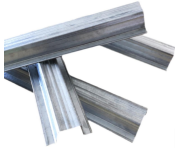
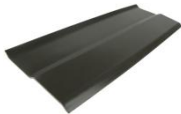
(2024·中英文版)

目录 CONTENTS

一、 材料清单 BILL OF MATERIAL	- 1 -
二、 工具准备 TOOLS	- 3 -
三、 安装顺序 INSTALLATIONS	- 4 -
四、 安装工序 INSTALLATION PROCESS	- 5 -
4.1 安装顺水条 INSTALL COUNTER BATTEN	- 5 -
4.1.1 工序简介 PROCESS INTRODUCTION	- 5 -
4.1.2 注意事项 PRECAUTION	- 5 -
4.1.3 工序材料 MATERIALS	- 5 -
4.1.4 使用工具 TOOLS	- 5 -
4.1.5 安装步骤 INSTALLATION STEPS	- 6 -
4.2 铺装复合型斜屋面防水卷材+铝箔复合隔热防水卷材 Paving composite sloping roof waterproofing membrane + aluminum foil composite thermal insulation waterproofing membrane	- 9 -
4.2.1 工序简介 PROCESS INTRODUCTION	- 9 -
4.2.2 注意事项 PRECAUTION	- 9 -
4.2.3 工序材料 MATERIALS	- 10 -
4.2.4 使用工具 TOOLS	- 10 -
4.2.5 安装步骤 INSTALLATION STEPS	- 10 -
4.3 安装挂瓦条 INSTALL THE TILE BATTENS	- 12 -
4.3.1 工序简介 PROCESS INTRODUCTION	- 12 -
4.3.2 注意事项 PRECAUTIONS	- 12 -
4.3.3 工序材料 MATERIALS	- 12 -
4.3.4 使用工具 TOOLS	- 12 -
4.3.5 安装步骤 INSTALLATION STEPS	- 13 -
4.4 安装封檐板 INSTALL FASCIA	- 15 -
4.4.1 工序简介 PROCESS INTRODUCTION	- 15 -
4.4.2 注意事项 PRECAUTION	- 15 -
4.4.3 工序材料 MATERIALS	- 16 -
4.4.4 使用工具 TOOLS	- 16 -
4.4.5 安装步骤 INSTALLATION STEPS	- 16 -
4.5 安装瓦片 INSTALL TILES	- 18 -
4.5.1 工序简介 PROCESS INTRODUCTION	- 18 -
4.5.2 注意事项 PRECAUTION	- 18 -
4.5.4 使用工具 TOOLS	- 19 -
4.5.5 安装步骤 INSTALLATION STEPS	- 20 -
4.6 节点收口 NODES(BOEDER LINES)	- 23 -
4.6.1 工序简介 PROCESS INTRODUCTION	- 23 -
4.6.2 边瓦收口 THE SIDE TILE	- 23 -
4.6.3 脊瓦收口 RIDGE TILE BORDER	- 25 -
4.6.4 泛水收口 FLASHING	- 28 -

一、材料清单 BILL OF MATERIAL

编号 NO	材料名称 NAME	规格 SPECIFICATION L*W	产品图 PHOTOS	使用部位 POSITIONS
1	Pro 发电瓦 SOLAR TILE	长*宽 1240*390 mm		坡屋面 SLOPE ROOF
2	Pro 非发电整瓦 TILE	1240*390		坡屋面 SLOPE ROOF
3	Pro 非发电 1/2 1/2 SIZE TILE	630*390		坡屋面 SLOPE ROOF
4	Pro 非发电 1/4 瓦 1/4 TILE	320*390		坡屋面 SLOPE ROOF
5	脊瓦 RIDGE TILE	165*390		屋脊 RIDGE
6	脊瓦封头 BLOCK-END RIDGE TILE	165*610		屋脊 RIDGE
7	边瓦 CLOAKED VERGE TILE	150*390		边瓦 BORDER TILE
8	边瓦封头 BLOCK-END BORDER TILE	150*390		边瓦 BORDER TILE
9	铝箔复合隔热防水垫层 ALUMINUM FOIL COMPOSITE INSULATION AND WATERPROOF CUSHION LAYER	单面银 SINGLE SIDED SILVER		斜天沟 TAPERED VALLEY
10	复合型斜屋面防 水卷材 WATERPROOF ROLL			屋面 ROOF
11	天沟板 GUTTER BOARD	——		斜天沟 VALLEY
12	封檐板 VERGE BOARD	——		檐口 CORNICE
13	丁基胶带 TAPE	15m/ROLL		檐口板 CORNICE BOARD

14	通长龙骨 STEEL BATTEN	30*30*1.5mm		屋脊 RIDGE
15	六角燕尾丝 HEXAGON SELF DRILLING SCREW	φ4.8*50mm		顺水条 COUNTER BATTEN
16	檐口板附板 SUB BOARD	——		檐口、山墙 CORNICE, WALL
17	沉头燕尾丝 SELF DRILLING SCREW	φ3.8*40mm		光伏瓦、配瓦 SOLAR TILE, TILE
18	抗风扣 WIND RESISTANT BUCKLE	——		光伏瓦固定位置 TILR FIX POSITION
19	顺水条 COUNTER BATTEN	67*30*21mm		屋面 ROOF
20	挂瓦条 TILE BATTEN	67*30*21mm		屋面 ROOF
21	普立得卷材 PLD COIL MATERIALS	——		泛水 FLASHING
22	墅封压条 EDGE SEALANT STRIP	——		泛水 FLASHING
23	锐固支架 BRACKET	——		屋脊 RIDGE
24	通风卷材 VENTILATED COIL MATERIALS	TYPE B WIDE 300		屋脊 RIDGE
25	耐候密封胶 WEATHER RESISTANT SEALANT			节点收口 BORDER
26	直流电缆 DC CABLE	PV1-F4		光伏瓦 SOLAR TILE
27	MC4 连接器 CONNECTOR			光伏瓦 SOLAR TILE

二、 工具准备 TOOLS

			
万用表 Multimeter	剥线钳 (压线钳) Wire Stripper	手枪钻 Pistol Drill	角磨机 Angle Grinder
			
切割锯 Cutting Saw	美工刀 Utility Knife	盒尺 Steel Tape	卷尺 Tape Measure
			
水平尺 Level Ruler	激光水平仪 Laser Level	锤子 Hammer	胶枪 Glue Gun
			
安全绳 Safety Rope	反光背心 Reflective Vest	安全帽 Helmet	护目镜 Glasses
			
扫帚 Bloom	墨斗 Ink Marker		

工具准备仅供参考，可根据本地情况进行可替代功能性采购相应工具，使用可根据安装步骤进行选择性使用，达到安装效果即可。

The TOOLS above are just for reference only.

三、 安装顺序 INSTALLATIONS

预先注意事项：

precautions

1、施工人员根据屋顶组件排布图纸熟悉整体排布，确定线缆进入屋顶的位置。

The constructors are familiar with the overall layout based on the roof component layout drawing, and determine the position of the cables entering the roof.

2、安装前请进行以下检查/Please perform the following checks before installation:

(1) 安装使用材料没有损坏，数量与材料清单一致。

Make sure there is no damaged material, and the material quantities are consistent with the lists.

(2) 安装工具准备齐全。

The TOOLS are fully prepared

(3) 安装前确保所有电缆连接器干燥，无污垢。

Make sure all cable connectors are dry and clean before installation

3、确认线缆连接方式。

Confirm the way of cable connection

将光伏瓦单元按照排布图串联成串，与预先铺设在屋顶上的光伏专用线缆连接，**线缆的另一侧与逆变设备连接。**

There solar units are connected in series according to the layout diagram, and connected with the special DC cable laid on the roof in advance, and then connect the other end of the cable to the inverter equipment.

序号 No.	安装方案名称 Installation Scheme	使用材料 Materials
1	安装顺水条 Install counter batten	67*30*21mm 镀铝锌 Metal batten φ4*8*50mm 六角燕尾丝 Drilling screw
2	铺装铝箔复合隔热防水垫层 Pave heat insulation waterproof underlayment	铝箔 Aluminum foil
3	安装挂瓦条 Install tile batten	67*30*21mm 镀铝锌 Metal batten φ4.8*50mm 六角燕尾丝 Drilling screw
4	安装封檐板 Install verge board	封檐板、丁基胶带、附板、φ3.8*40mm 沉头燕尾丝 Verge board, tape, board, drilling screw
5	安装斜天沟 Install tapered valley	天沟板 Board
6	安装瓦片 Install tiles	光伏瓦、配瓦、沉头燕尾丝、抗风扣、直流电缆、MC4 连接器、光伏排布图 Solar tiles, tiles, screw, buckle DC cable, connector, solar tile layout diagram
7	节点收口 Border	脊瓦、边瓦、墅封压条、耐候密封胶 Ridge tile, border tile, Edge sealant strip, sealant
8	安装电气材料 Install electrical materials	屋顶组串图 Diagram

四、 安装工序 INSTALLATION PROCESS

4.1 安装顺水条 INSTALL COUNTER BATTEN

4.1.1 工序简介 PROCESS INTRODUCTION

顺水条即与挂瓦条垂直相交的木条，用来固定挂瓦条、架空屋面瓦下面，有利于屋顶通风。

The counter battens crossing vertically with the tile battens are used to fix tile battens on them. The free space under the tiles is conducive to roof ventilation.

4.1.2 注意事项 PRECAUTION

施工面要求：

Construction surface requirements

1、混凝土基层面：扫把清扫，不得有油污、突起物、裂缝、蜂窝、麻面、起皮和起砂现象；

Concrete base surface: Sweep with a broom, there should be no oil stains, protrusions, cracks, honeycombs, rough surfaces, peeling, or sanding;

2、伸出屋面的管道、设备或预埋件等，应在附加防水层施工前安装完毕；

Pipes, equipment, or embedded parts extending from the roof should be installed before the construction of the additional waterproof layer

3、要求施工面平整度误差小于等于 40mm，平整度误差大于 40mm，采用防腐木龙骨找平。满足上述条件，方可进行下一道工序的施工。

Require the flatness error of the construction surface to be no more than 40mm, and if the flatness error is greater than 40mm, use anti-corrosion wooden keel for leveling.

Only when the above conditions are met can the construction of the next process be proceed.

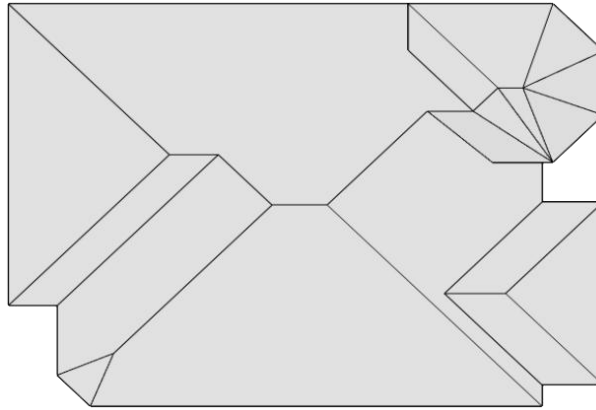
4.1.3 工序材料 MATERIALS

	
顺水条 counter batten 67*30*21mm	六角燕尾丝 screw φ4.8*50mm

4.1.4 使用工具 TOOLS

			
手枪钻 pistol drill	切割锯 cutting saw	激光水平仪 laser level	墨斗 ink marker
			
盒尺 steel tape	卷尺 tape measure	水平尺 level ruler	

4.1.5 安装步骤 INSTALLATION STEPS



基层示意图

Schematic diagram of the base layer

1、 在基层的斜脊、天沟位置，增设顺水条。

At the position of the inclined ridge and gutter of the base layer, add counter battens.

A、使用激光水平仪进行测量水平位置。

Use a laser level to measure the horizontal positions.

B、使用墨斗进行弹线。

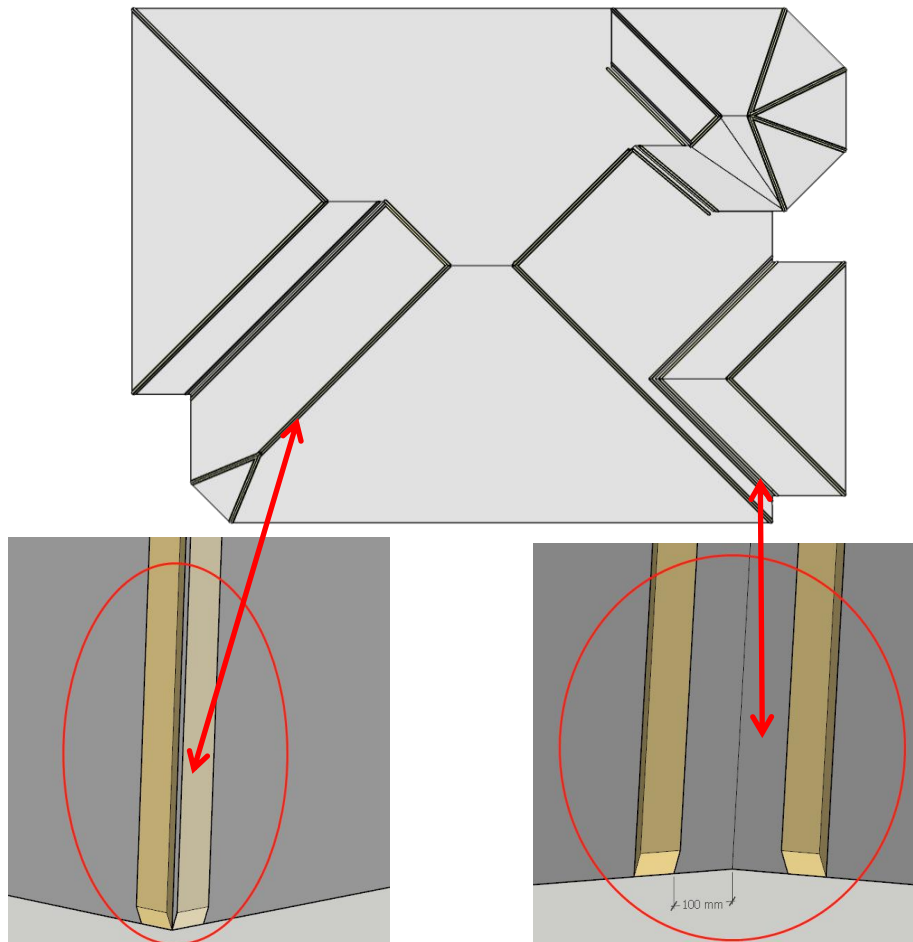
Draw lines with an ink marker.

C、使用盒尺标记距离中线 $\geq 100\text{mm}$ 位置， $\phi 4.8 \times 50\text{mm}$ 六角燕尾丝固定间距 $\leq 400\text{mm}$ 。

Use a tape measure to mark the position at a distance of $\geq 100\text{mm}$ from the centerline, with a fixed spacing of $\leq 400\text{mm}$ for hexagonal drill screws of $\phi 4.8 \times 50\text{mm}$.

D、在斜脊、天沟位置放置顺水条，使用手枪钻将沉头燕尾丝穿过顺水条固定在屋面基层上，依此类推，直至整个屋顶斜脊和天沟位置铺设完成。

Place counter battens at the ridge and gutter positions, use a pistol drill to thread the countersunk drilling screw through the battens and fix them on the roof base, until the entire roof ridge and gutter positions are laid.



斜脊两侧顺水条安装位置需和斜脊中线
保持一致

The counter battens on the both sides of
the inclined ridge should be consistent
with the center line of the inclined ridge

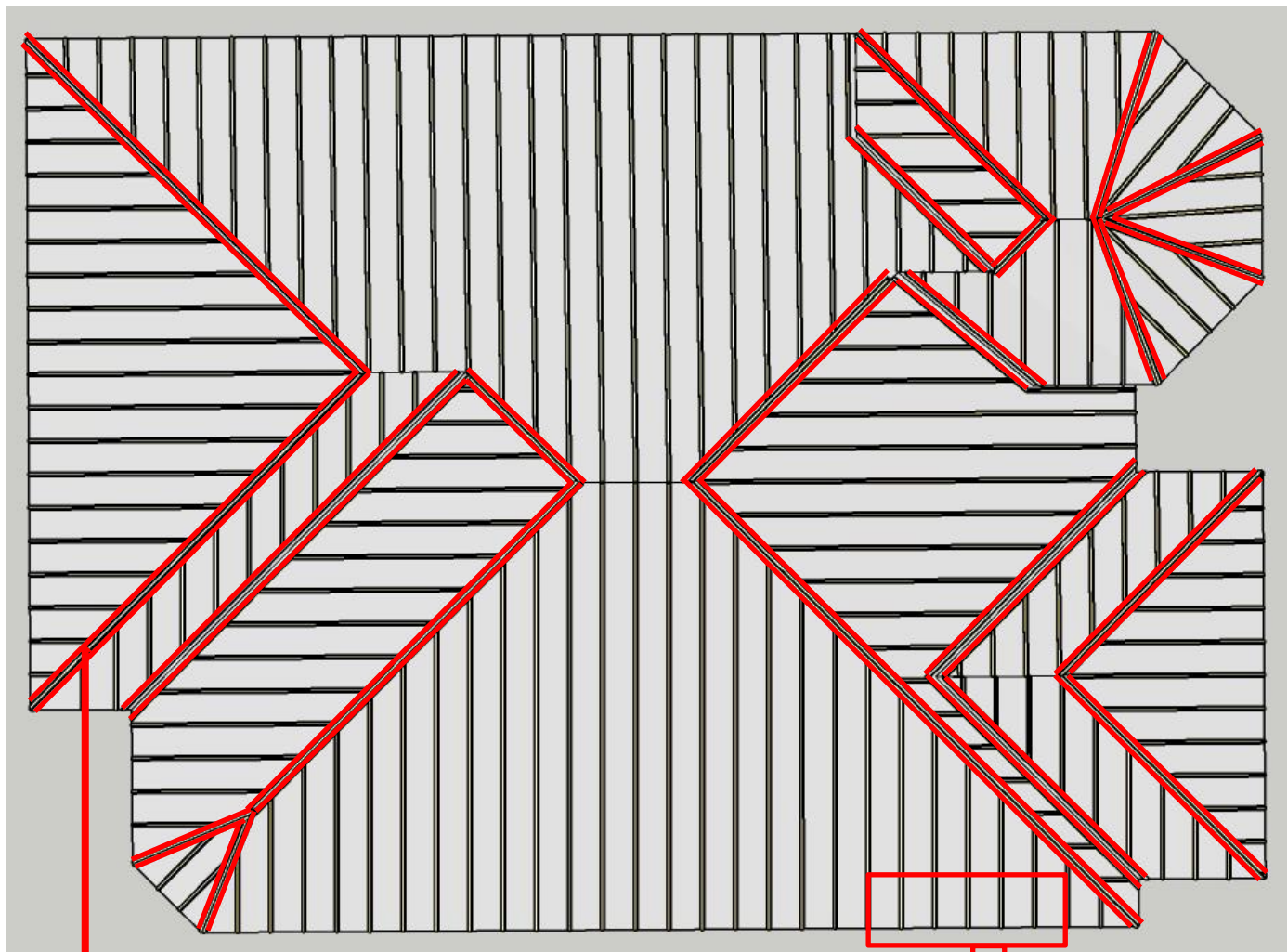
天沟两侧顺水条安装位置需和基层天沟
中线距离 $\geq 100\text{mm}$

The battens on both sides of the gutter
should be $\geq 100\text{mm}$ away from the
center line of the base gutter

2、主屋面位置安装顺水条；

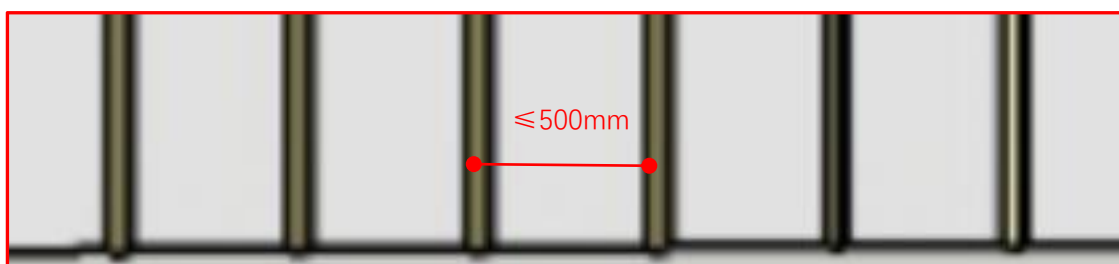
Install counter battens on the main roof

- A、使用激光水平仪进行测量水平位置。
Use a laser level to measure the horizontal positions.
- B、使用墨斗进行弹线。
Draw lines with an ink marker.
- C、使用盒尺标记顺水条间距 $\leq 500\text{mm}$ ，沉头燕尾丝固定间距 $\leq 400\text{mm}$ 。
Use a tape measure to mark the position at a distance of ≤ 500 between the battens,
and a fixed spacing of $\leq 400\text{mm}$ for sunkhead drill screws.
- D、依此将顺水条垂直于檐口均匀放置，使用手枪钻将沉头燕尾丝穿过顺水条固定在屋面基
层上，依此类推，施工直至整个屋面铺设完成。
Place the battens evenly vertical to the cornice, use a pistol drill to thread the
countersunk drilling screws through the battens and fix them on the roof base, until the
entire roof is laid.



上图红色标记位置为增设顺水条，安装方式见上一页。

The red lines are the add counter battens. See the previous page for the installation.



其余位置顺水条布置图

Layout of the rest counter battens

4.2 铺装复合型斜屋面防水卷材+铝箔复合隔热防水卷材 Paving composite sloping roof waterproofing membrane + aluminum foil composite thermal insulation waterproofing membrane

4.2.1 工序简介 PROCESS INTRODUCTION

该材料隔热性能卓越，80%的热射线反射率，有效起到室外隔热及室内保温的双向效果，同时满铺后具有良好的防水性能，且产品抗老化性能高，柔软轻薄，易于施工无死角。

These materials have excellent thermal insulation performance, 80% of the thermal ray reflectivity, effectively playing a two-way effect of outdoor insulation and indoor insulation. At the same time, they have good waterproof performance after full paving, and the products have high anti-aging performance, soft and thin, easy to construct without dead corners.

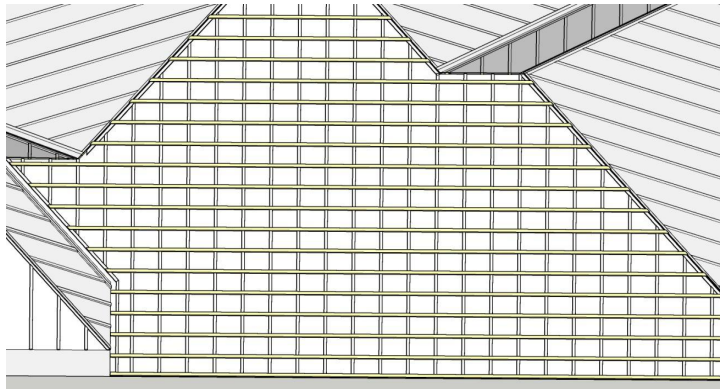
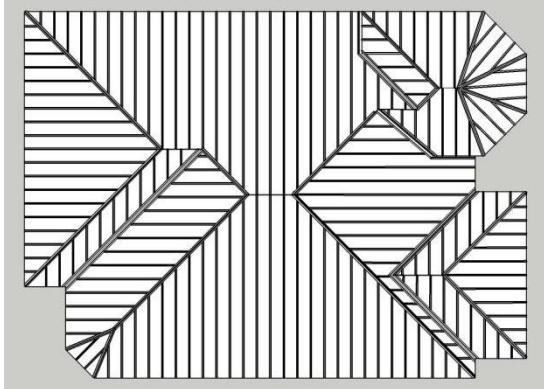
4.2.2 注意事项 PRECAUTION

- 1、顺水条铺设完成后铺装复合型斜屋面防水卷材+铝箔复合隔热防水卷材，

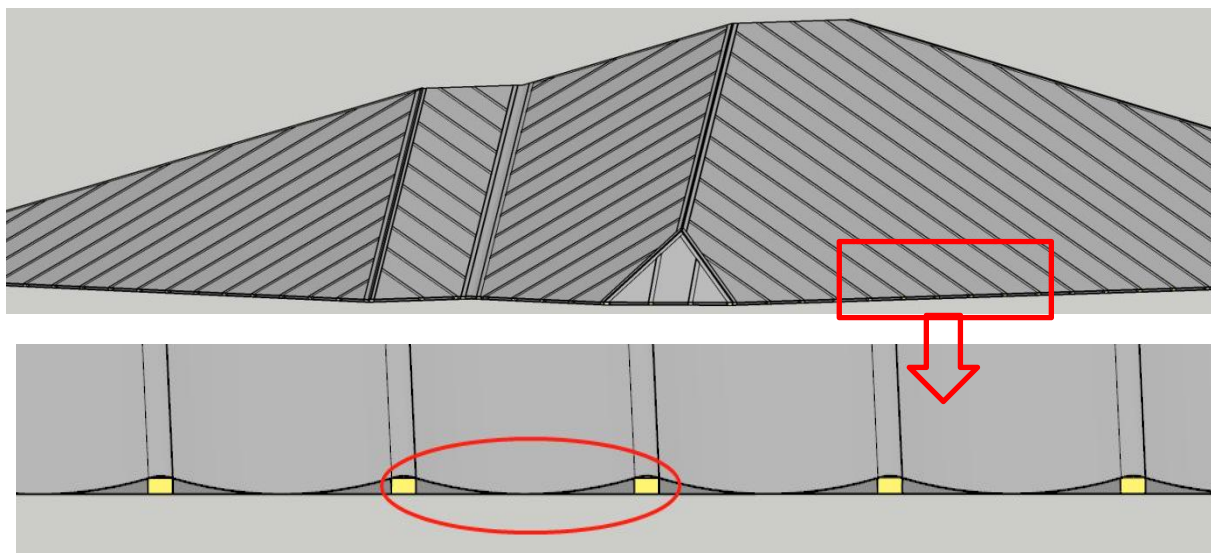
After the completion of installing the counter battens, a composite sloping roof waterproofing membrane and an aluminum foil composite thermal insulation waterproofing membrane can be paved,

- 2、复合型斜屋面防水卷材+铝箔复合隔热防水垫层铺设一层，使用挂瓦条进行固定安装（安装步骤见 4.3），4.2 和 4.3 两个工序必须同步进行安装，依此类推，屋面安装完成如下图。

A layer of composite inclined roof waterproofing membrane and aluminum foil composite thermal insulation waterproof cushion layer is laid, and fixed with tile battens (see 4.3 for installation steps). The two installation processes of 4.2 and 4.3 must be done simultaneously. The roof installation is completed as shown in the following figure.



- 1、正脊、斜脊处搭接时，搭接位置不能出现在屋脊正上方，必须在距离屋脊 $\geq 500\text{mm}$ 处进行上下搭接。
When lapping is at the ridge and the inclined ridge, the overlap position cannot appear directly above the ridge, and the upper and lower lap must be carried out at a $\geq$ distance of 500mm from the ridge.
- 2、顺水条之间应呈凹状(见附图)
The shape between the counter battens should be concave (see attached picture)



4.2.3 工序材料 MATERIALS

	
铝箔复合隔热防水垫层 ALUMINUM FOIL COMPOSITE INSULATION AND WATERPROOF CUSHION LAYER 单面银 single side silver	六角燕尾丝 drilling screws φ4.8*50mm

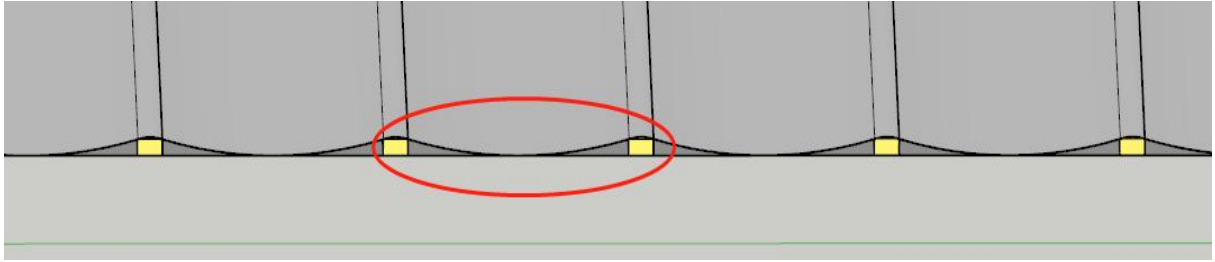
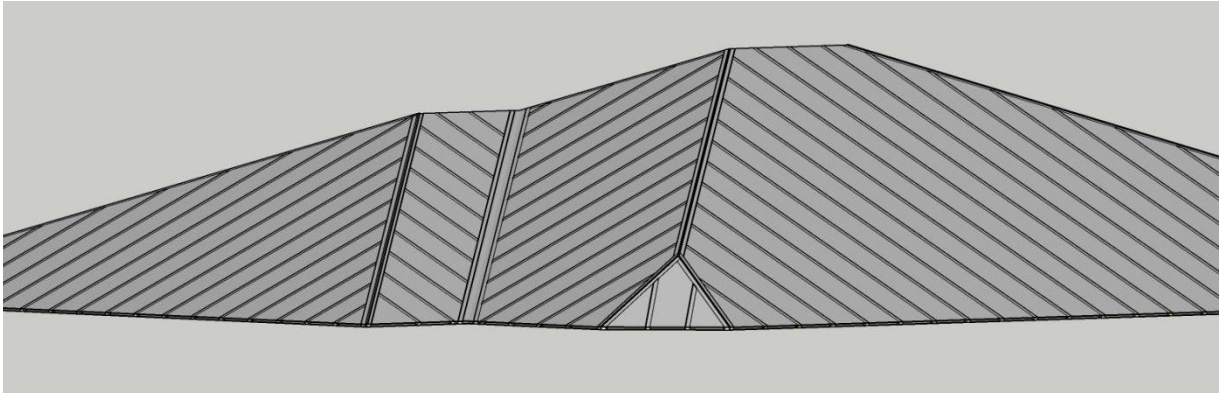
4.2.4 使用工具 TOOLS

		
手枪钻 pistol drill	盒尺 steel tape	美工刀 utility knife

4.2.5 安装步骤 INSTALLATION STEPS

1、在顺水条上方，由屋面檐口位置开始（建议从单坡面积较大的屋面开始），由下至上进行水平横向铺设复合型斜屋面防水卷材+铝箔复合隔热防水垫层，上下搭接重叠尺寸 $\geq 100\text{mm}$ 。依此类推，直至整个屋面铺设完成。

Starting from the position of the cornices above the counter batten (it is recommended to start from the roof with a larger single slope area), horizontally lay composite inclined roof waterproofing membrane and aluminum foil composite thermal insulation waterproof cushion layer from bottom to top, with an overlap size of $\geq 100\text{mm}$. Repeat this process until the entire roof is laid.



4.3 安装挂瓦条 INSTALL THE TILE BATTENS

4.3.1 工序简介 PROCESS INTRODUCTION

挂瓦条指在坡屋面上排列瓦片的工序方法，一般在屋面安装挂瓦条，将瓦挂在挂瓦条上。

This process is to install tile battens on the roof. The tiles are fixed on the tile battens.

4.3.2 注意事项 PRECAUTIONS

1、挂瓦条应安装平直，边施工边检查，随时纠正偏差以免返工；

The tile battens should be installed straight, checked during the process of installation, and any deviations should be corrected at any time to avoid rework

2、挂瓦条长度不够时直接对缝拼接，超出时切掉多余长度；

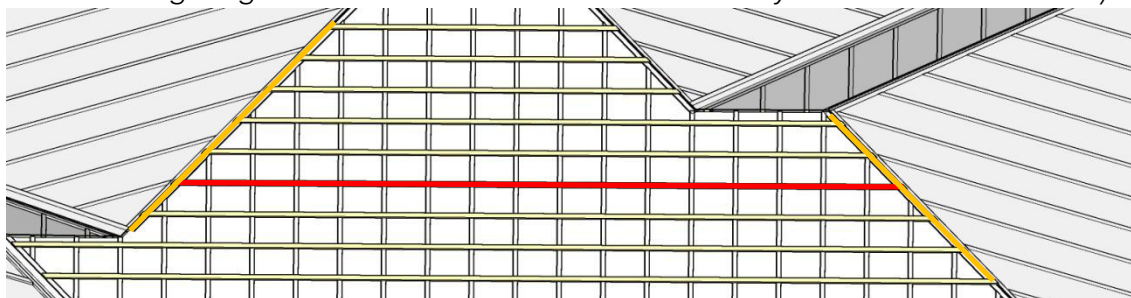
If the length of the tile batten is not enough, it can be directly spliced together. If it exceeds the limit, the excess length should be cut off

3、固定钉应完全拧入挂瓦条，钉头不得突出挂瓦条表面；

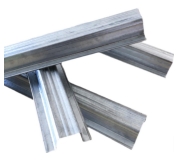
The fixing screws should be completely screwed into the hanging tile, and the screw head should not protrude from the surface of the batten.

4、挂瓦条端部不可超过附加顺水条边界（如下图标记：红线两端不可超过黄线）

The end of the batten should not exceed the boundary of the additional counter battens (as shown in the following diagram: the red line shall not exceed the yellow line at both ends)



4.3.3 工序材料 MATERIALS

	
挂瓦条 tile batten 67*30*21mm	六角燕尾丝 drilling screw φ4.8*50mm
见材料清单 编号 24 Refer to Material List No. 24	见材料清单 编号 25 Refer to Material List No. 25

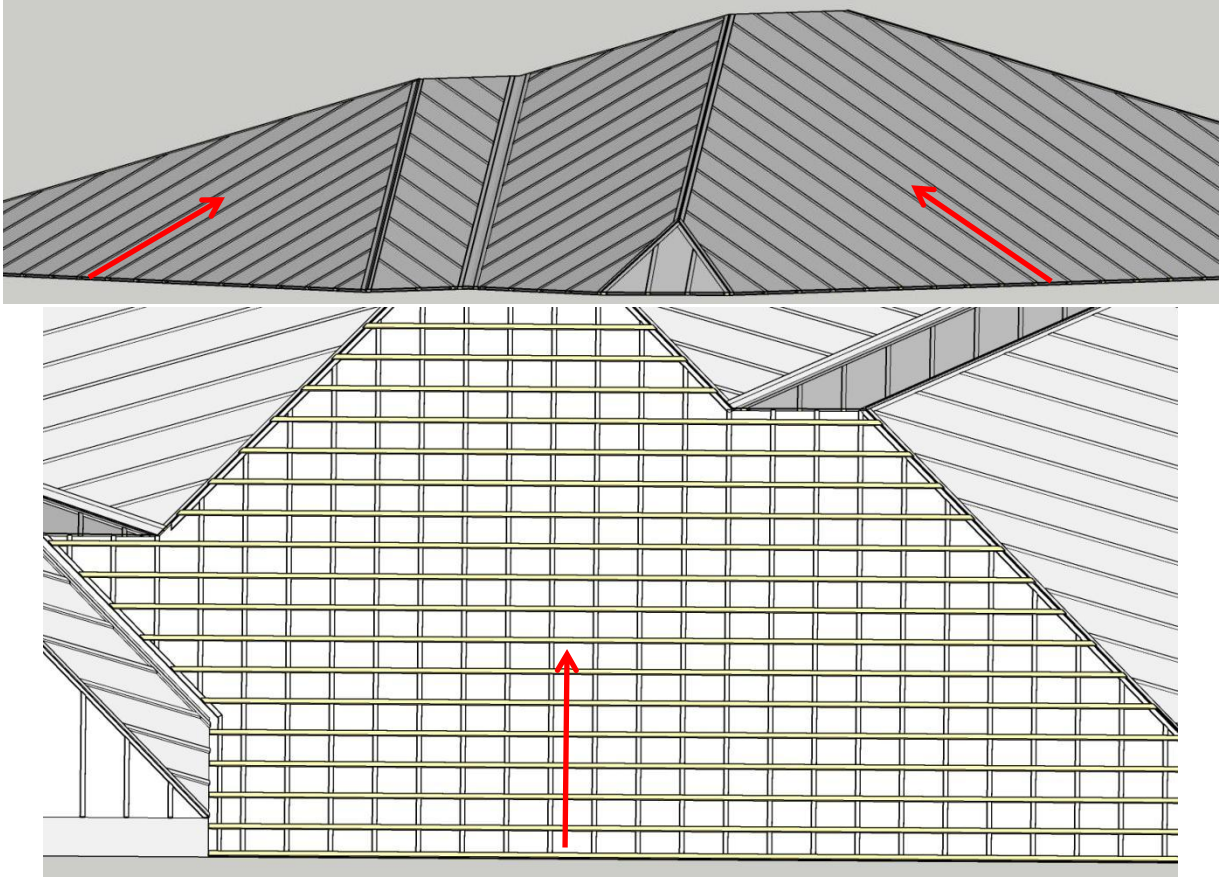
4.3.4 使用工具 TOOLS

			
手枪钻 pistol drill	切割锯 cutting saw	盒尺 steel tape	墨斗

4.3.5 安装步骤 INSTALLATION STEPS

1、挂瓦条自下而上逐排水平排布，按照规定间距进行安装固定。

The tile battens are arranged horizontally from bottom to top, and installed and fixed according to the specified spacing.



2、安装挂瓦条 install the tile battens

A、最底部第一排挂瓦条作为封檐板安装条使用，从第二排挂瓦条开始作为瓦片安装条使用。

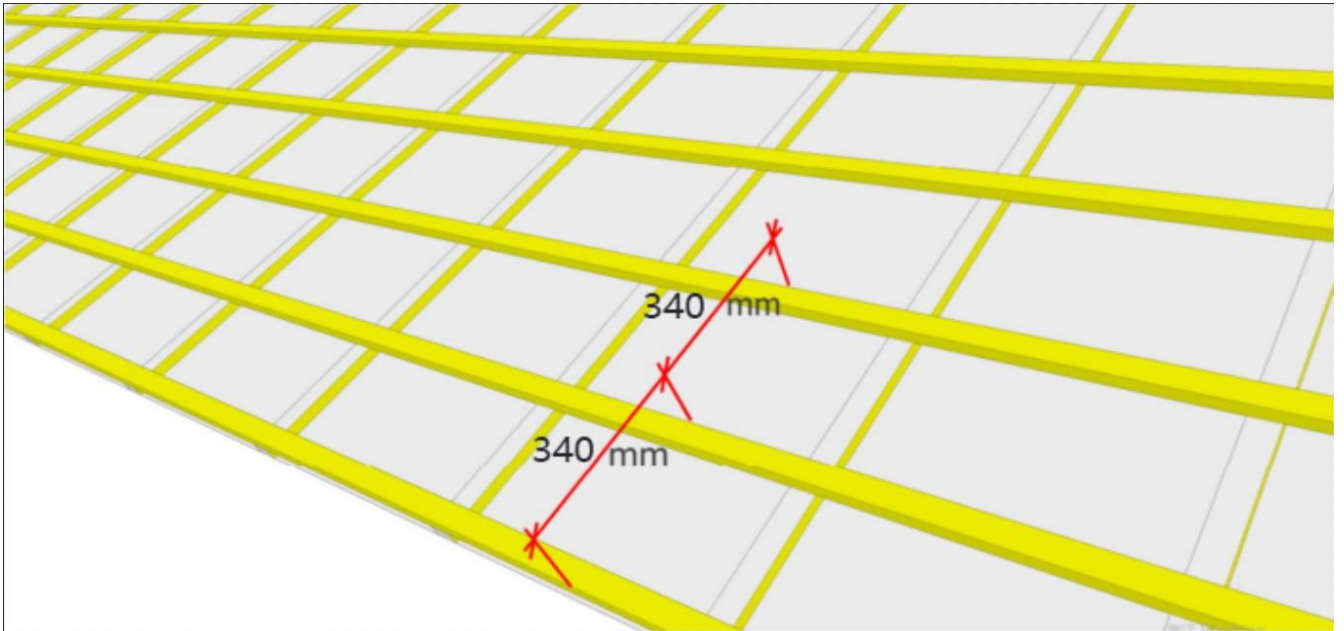
The first row of batten at the bottom is used as the batten for fascia mounting, and from the second row of battens are used for the tile mounting.

B、第一排挂瓦条下方与檐口墙体水平对齐，第二排与第一排间距为 340mm，第三排与第二排间距为 340mm，第三排往上均按照间距 340mm 进行安装，直至屋面顶部屋脊处。

Align the lower part of the first row of batten horizontally with the cornice wall, the distance between the second row and the first row is 340mm, the distance between the third row and the second row is 340mm, and the installation of the third row upwards is carried out according to the distance of 340mm until the top ridge of the roof is reached

C、挂瓦条与顺水条的交叉点均为固定点，使用沉头燕尾丝固定。

The intersection points of the battens are fixed points, fixed the battens with countersunk drilling screws.



4.4 安装封檐板 INSTALL FASCIA

4.4.1 工序简介 PROCESS INTRODUCTION

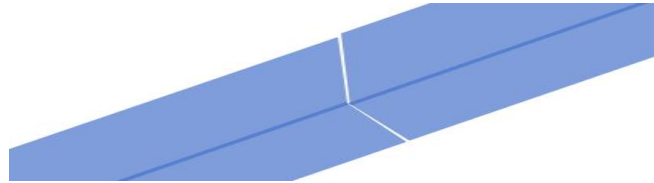
封檐板是指钉置在坡屋顶檐口外侧挑檐处的金属板，增加了檐口位置的美观及防水性。

The fascia (cornice board) refers to the metal plate fixed to the outside of the cornice of the sloping roof, which increases the beauty and waterproofness of the cornice position.

4.4.2 注意事项 PRECAUTION

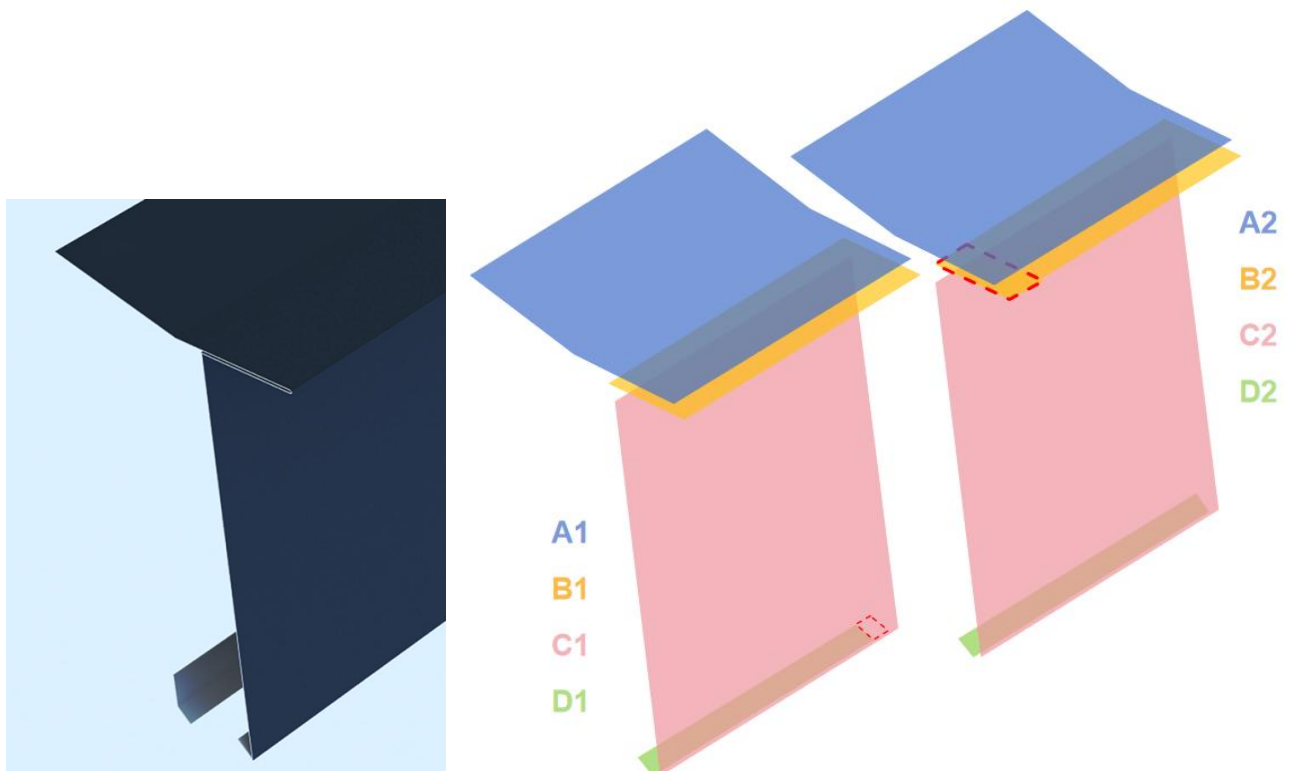
1、附板安装不需要左右搭接，直接对缝安装。如下图

The installation of the attachment plate does not require overlap, and they can be directly installed directly against the seam. As shown in the figure below

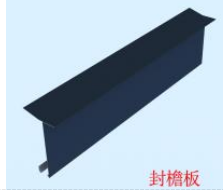
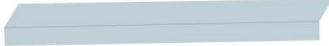


2、封檐板由屋面檐口右侧向左侧安装，左侧封檐板在上方，右侧封檐板在下方，左右搭接安装，搭接时，搭接侧口 B2、底口 D1 剪裁后再搭接，A2 和 B2 面链接弯曲处需剪裁掉，保留 A2 面即可，如不裁剪会影响搭接，剪裁的目的是保证搭接重叠部分能够更紧密，固定更牢固。剪裁宽度为 50mm，搭接间距 50mm，裁剪方式见下图：

The fascia is installed from the right side to the left side, with the left fascia above and the right fascia below. They are installed by overlapping left and right. When overlapping, the overlapping side opening B2 and bottom opening D1 are cut and then overlapped. The curved connection between A2 and B2 needs to be cut off, and A2 can be retained. If not cut, it will affect the overlapping. The purpose of cutting is to ensure that the overlapping part can be more tightly and firmly fixed. The cutting width is 50mm, and the overlap spacing is 50mm. The cutting method is shown in the following figure:



4.4.3 工序材料 MATERIALS

			
封檐板 fascia	附板 attachment plate	丁基胶带 BUTYL TAPE	沉头燕尾丝 drill screw Φ3.8*40mm

4.4.4 使用工具 TOOLS

			
手枪钻 pistol drill	切割锯 cutting saw	角磨机 angle grinder	盒尺 steel tape
			
美工刀	铁皮电剪 Tin electric scissors	铁皮剪 Tin scissors	水平尺 level ruler

4.4.5 安装步骤 INSTALLATION STEPS

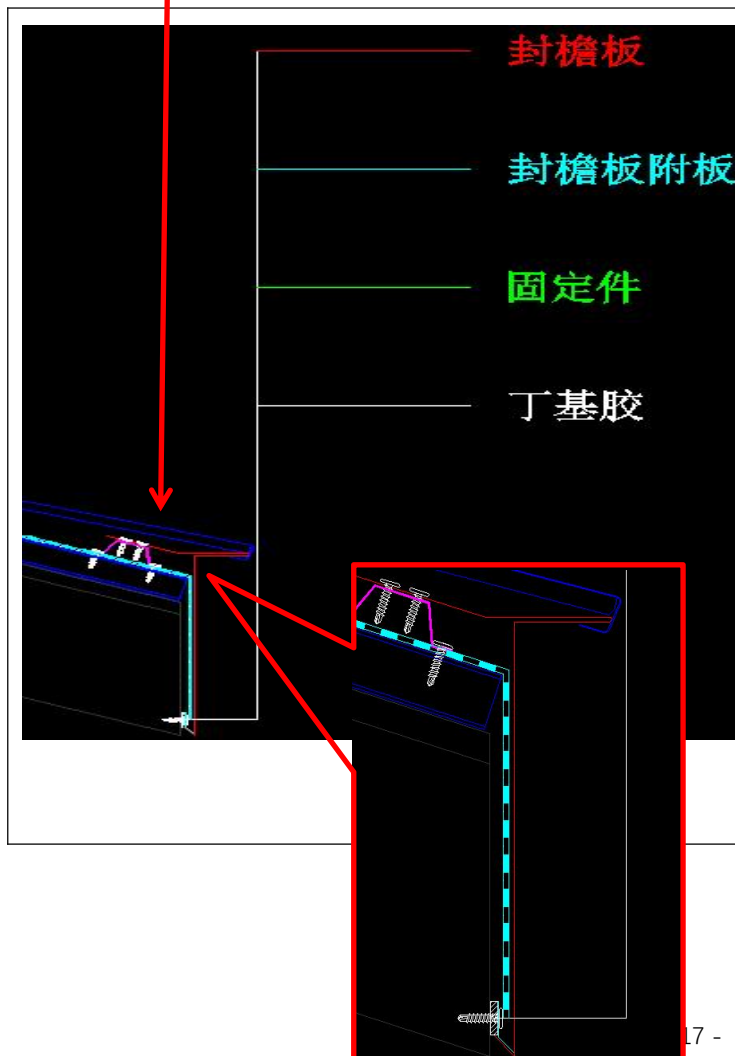
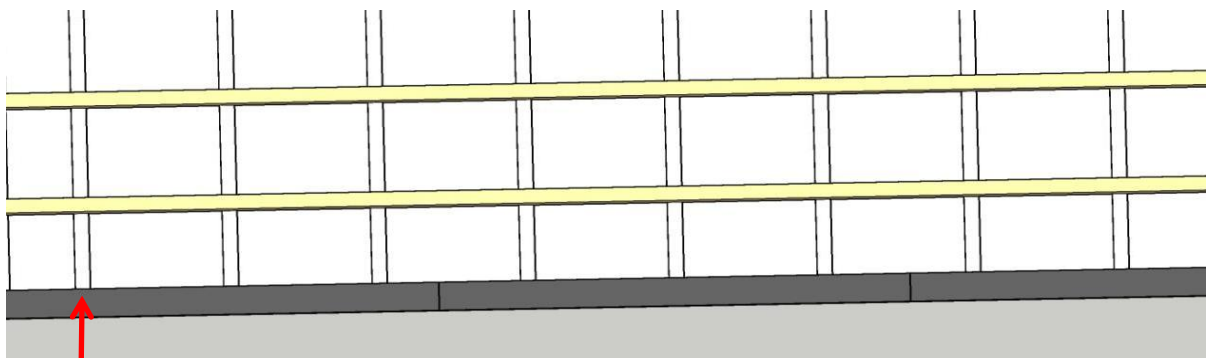
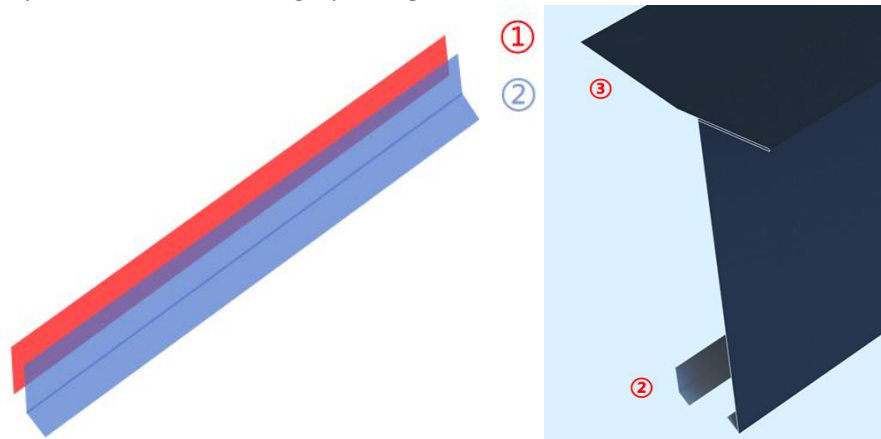
- 1、裁剪与附板同样长度的丁基胶带贴在附板背部；如下图：①是丁基胶带，②是附板，
Cut butyl tape of the same length as the attached board and stick it on the back of the attached board; As shown in the following figure: ①: butyl tape, ②: attachment board,



- 2、将封檐板放置在第一排挂瓦条上，确定附板在檐口的立面墙体位置；
Place the fascia on the first row of tile batten, and determine the position of the attached board on the facade wall of the cornice.
- 3、按照确定附板位置，水平依次布置附板；
Arrange the attachment plates horizontally in sequence according to the determined attachment plate position.
- 4、丁基胶带位于墙体于附板之间，使用沉头燕尾丝直接穿透附板固定在墙体上，螺钉固定间距 $\leq 400\text{mm}$ ；
Butyl tape is located between the wall and the attachment plate, and is fixed to the wall directly by penetrating the attachment plate with countersunk drilling screws, with a screw fixing spacing of $\leq 400\text{mm}$;
- 5、沿第一排挂瓦条方向依次由屋面檐口右侧向左侧放置封檐板，底部勾住附板，上部用沉头燕

尾丝穿透封檐板固定在挂瓦条上，螺钉固定间距 $\leq 400\text{mm}$

Place the faascias from the right side to the left side along the direction of the first row of tile batten, hook the attached panel at the bottom, and fix the fascia on the tile batten with countersunk drilling screw at the top, with a screw fixing spacing of $\leq 400\text{mm}$

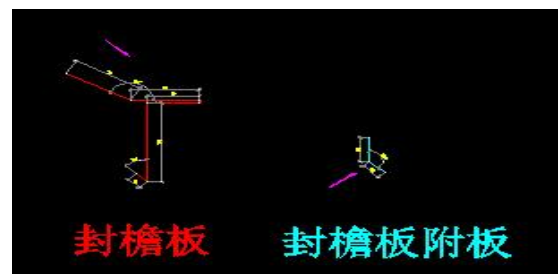


封檐板 VERGE BOARD

封檐板附板 SUB BOARD

固定件 DRILL SCREW

丁基胶带 BUTYL TAPE



封檐板
VERGE BOARD

附板
SUB BOARD

4.5 安装瓦片 INSTALL TILES

4.5.1 工序简介 PROCESS INTRODUCTION

光伏瓦及金属瓦安装说明

installation instruction for solar tiles and metal tiles.

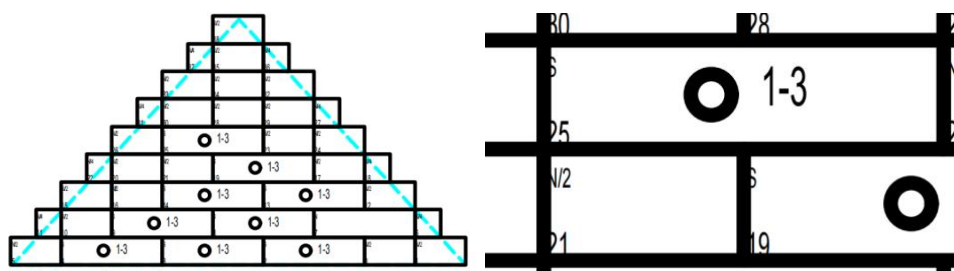
4.5.2 注意事项 PRECAUTION

1、安装前根据现场并结合屋顶，确定线缆在屋顶的位置，预先在屋顶上铺设光伏专用线缆。

Before installation, determine the location of the cable on the roof according to the site and combined with the roof, and lay the special cable on the roof in advance.

2、光伏瓦布置图中标识说明：S 代表光伏瓦，N 代表切割配瓦，

In the layout diagram of the tiles, S represents solar tiles, N represents cutting and matching tiles,



3、安装光伏瓦时，应随时用万用表测试光伏瓦与光伏瓦之间连接是否有效，确保通电，每个区域串联之后，使用万用表测试通路，确保通电。光伏瓦排布完成之后，开始安装排线，所有线缆均在瓦片下方。

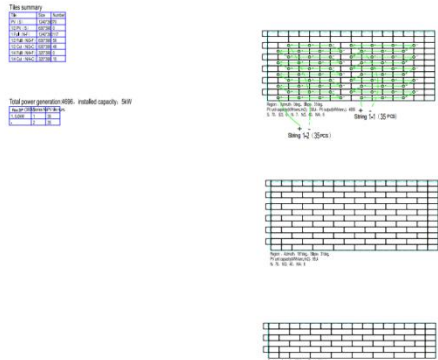
While installing solar tiles, a multimeter should be used at all times to test whether the connections between the solar tiles are effective and ensure power supply. After connecting each area in series, a multimeter should be used to test the circuit and ensure power supply. After the layout of all the solar tiles is completed, the installation of the cables begins, and all cables are located under the tiles.

4、屋面顶部到正脊、两侧到山檐、斜天沟斜脊的边部位置使用切割。

Cutting tiles are used for the edge positions from the top of the roof to the main ridge, both sides to the eaves, and the sloped ridge of the gutter.

5、每个屋面的瓦片由非发电配瓦和光伏瓦或只有非发电配瓦组成，安装时请严格参照本节“节点安装瓦片说明”中的情形进行操作，避免返工。Each roof tile is composed of ordinary tiles and solar tiles or only ordinary tiles. Please strictly follow the instructions in this section's "Installation Tile Instructions" during installation to avoid rework.

4.5.3 工序材料 materials

				
PRO 发电瓦 solar tile	Pro 非发电 1/2 瓦 Ordinary tile	Pro 非发电 1/4 瓦 Ordinary tile	抗风扣 Wind resistant buckle	沉头燕尾丝 $\Phi 3.8 \times 40\text{mm}$ Drilling screw
				
直流电缆 PV1-F4 DC cable	MC4 连接器 connector			
			光伏排布图 solar tile layout diagram	

4.5.4 使用工具 TOOLS

				
手枪钻 pistol drill	切割锯 cutting saw	角磨机 corner grinder	盒尺 steel tape	激光水平仪
				
万用表 multimeter	剥线钳 (压线钳) wire strippers			

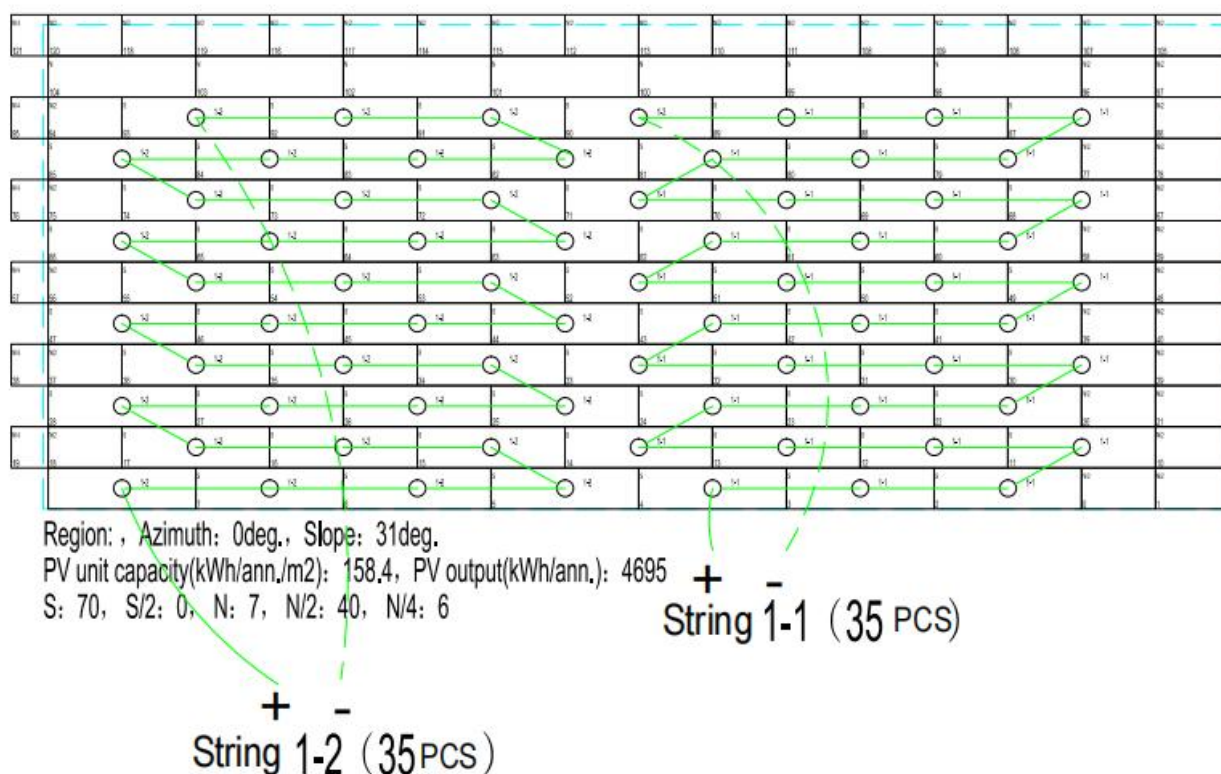
4.5.5 安装步骤 INSTALLATION STEPS

1、瓦片从每个屋面的左下角开始排布，逐片安装固定。

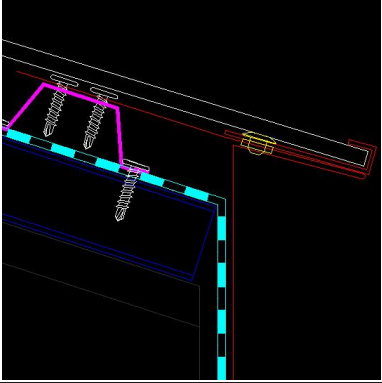
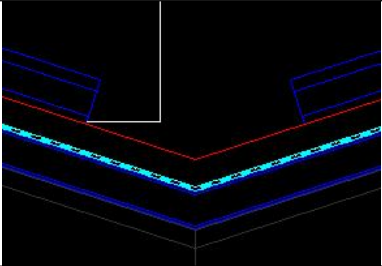
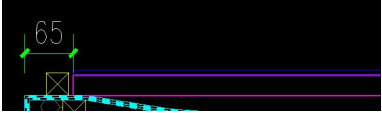
The tiles are fixed from the lower left corner of each roof, and are installed and fixed one by one;

2、按照每个户型的排布图进行瓦片排布。根据排布图标识（S（○）代表光伏瓦，N 非发电配瓦）选择相对应的瓦片。

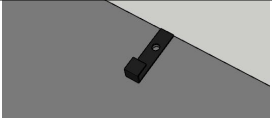
Install tiles according to the layout plan of each unit. According to the layout diagram identification Select the corresponding tile (S (○) representing solar tiles, N ordinary tiles).



3、节点安装瓦片说明：Border line Installation Tile Description:

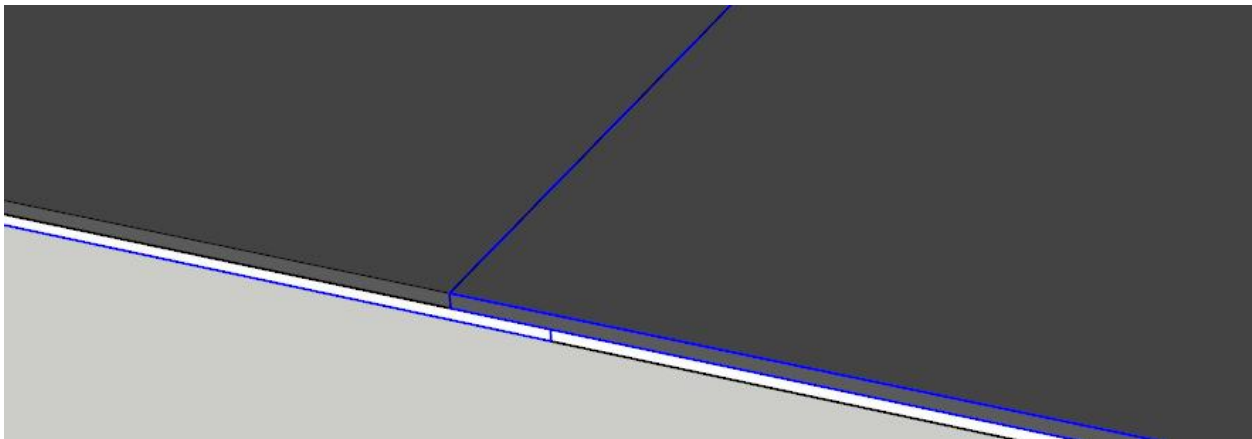
光伏瓦与封檐板连接处： The junction between the solar tile and the fascia 在封檐板上安装抑风扣，黑色拉铆钉固定 (图示：蓝色为金属瓦，红色为封檐板) Install the wind resistant buckle on the fascia and fix it with black rivets (illustration: blue is metal tile, red is fascia)	
斜天沟 Tapered valley	
屋脊 Ridge	安装至屋面界限即可。 需要将超出界限部分裁切整齐。 Install to the roof boundary. Need to cut the parts that exceed the boundary neatly.
边瓦 edge tile 主瓦安装至边瓦位置时，需要与边界线留出$\geq 65\text{mm}$空隙，靠近主瓦除沿坡度方向通长固定边瓦龙骨 When installing the main tile to the edge tile position, it is necessary to leave a gap of $\geq 65\text{mm}$ with the boundary line, and fix the edge tile keel along the slope direction near the main tile.	

4、瓦片固定材料及用途说明：Tile fixing materials and use description

瓦片类型 Tile type	固定材料 fixmaterila	固定位置 Fixing position	抗风扣 Wind resistant buckle
光伏瓦 soalr tile	固定钉使用 沉头燕尾丝 Fixing drilling screw $\Phi 3.8 \times 40\text{mm}$	瓦片上预留的固定孔 Reserved fixing holes on the tiles	 每片光伏瓦使用 3 个抗风扣 每个抗风扣使用 1 个固定钉 Each tile uses 3 wind resistant buckles. 1 fixing nail for each buckle

5、光伏瓦自带左右搭接结构，直接拼装

The solar tile comes with a left and right overlapping structure, which can be directly assembled.



6、光伏瓦上下搭接：上排光伏瓦底部插入下排抗风扣凹槽处

Solar tile upper and lower lap: the bottom of the upper row of solar tiles is inserted into the groove of the lower row of wind resistance buckles



4.6 节点收口 NODES(BOEDER LINES)

4.6.1 工序简介 PROCESS INTRODUCTION

节点即建筑专业节点，包括：檐口、檐沟、女儿墙、泛水、屋脊、屋面设施基础及管道伸出屋面处理等，

Nodes refer to architectural professional nodes, including fascia, gutters, parapets, flashing, ridges, roof infrastructure foundations, and pipe extensions for roof treatment.

“收口”也叫“收边”，是通过对装饰面的边、角以及衔接部分的工艺处理，以达到弥补饰面装修的不足之处，增加装饰效果的目的。收口是装修工程中细部工程的重要一环，能否做好收口关系到装修的最终效果。

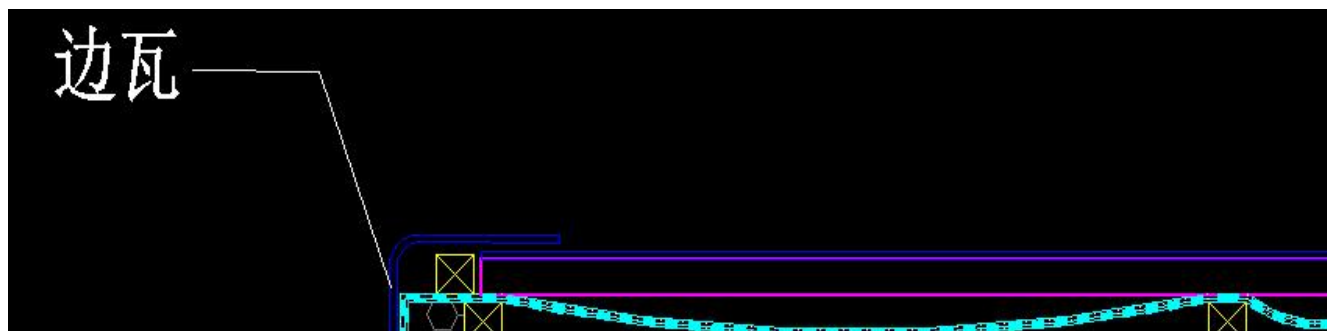
Closing ", also known as "border", is the process of treating the edges, corners, and connecting parts of the decorative surface to make up for the shortcomings of the decorative finish and increase the decorative effect. Closing the border is an important part of the detailed process in decoration projects, and whether it can be done well is related to the final effect of decoration.

4.6.2 边瓦收口 THE SIDE TILE

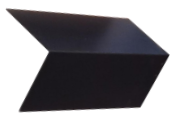
4.6.2.1 注意事项 PRECAUTION

1、在主瓦与山檐的空隙处沿坡度方向固定边瓦龙骨

The edge tile keel is fixed along the slope direction at the gap between the main tile and the cornice.



4.6.2.2 工序材料 MATERIALS

			
边瓦 edge tile	边瓦封头 edge tile head	通长龙骨 Steel batten 30*15*1.5mm	沉头燕尾丝 Drilling screw Φ3.8*40mm

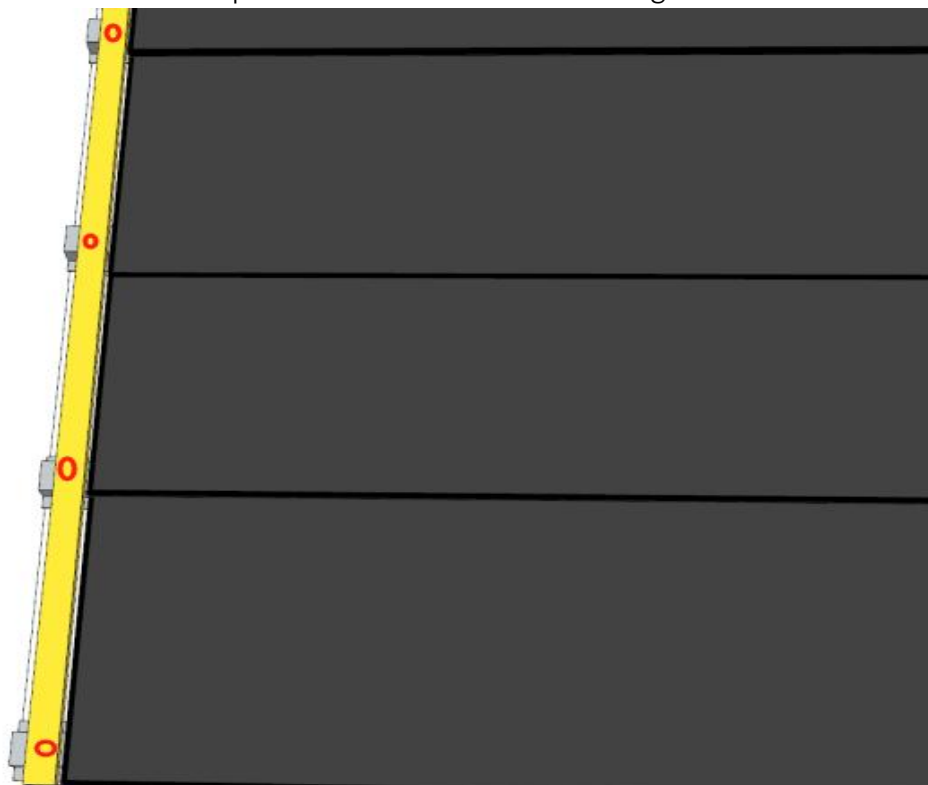
4.6.2.3 使用工具 TOOLS

	
手枪钻 pistol drill	切割锯 cutting saw

4.6.2.4 安装步骤 INSTALLATION STEPS

1、沿坡度方向固定边瓦通长龙骨，如下图方式布置并用 $\phi 4.8 \times 50\text{mm}$ 沉头燕尾丝固定

Fix the edge tile along the slope direction and the long keel, which is arranged as shown in the following figure and fixed with $\phi 4.8 \times 50\text{mm}$ countersunk drilling screws



2、边瓦由低到高依次安装：

The edge tiles are installed sequentially from low position to high

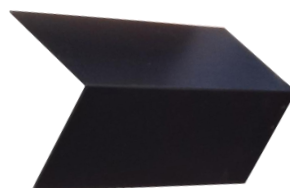
A、山檐底部安装边瓦封头， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝固定在通长龙骨上

Install edge tile heads at the bottom of the cornice, and fix them with a countersunk drilling screw on the full-length keel



B、连续布置边瓦到顶部， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝固定在通长龙骨上，边瓦两端尺寸不等，尺寸较大朝下扣住前一片

Continuously arrange the edge tiles to the top, fix them on the full-length keel with a countersunk drilling screws $3.8 \times 40\text{mm}$, and cover the front piece downwards with the larger tiles of different sizes at both ends





4.6.3 脊瓦收口 RIDGE TILE BORDER

4.6.3.1 注意事项 PRECAUTION

1、脊瓦与两侧的主瓦搭接距离 $\geq 40\text{mm}$

The lap distance between the ridge tile and the main tile on both sides $\geq 40\text{mm}$

	通长龙骨 Steel batten
	锐固支架 brfacket
	脊瓦 ridge tile
	通风卷材 Ventilation membranes

4.6.3.2 工序材料 MATERIALS

脊瓦 ridge tile	脊瓦封头 ridge tile head	锐固支架 brfacket	通长龙骨 Steel batten	通风卷材 Ventilation membranes	沉头燕尾丝 $\Phi 3.8 \times 40\text{mm}$

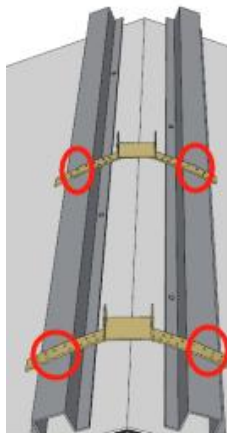
4.6.3.3 使用工具 TOOLS

手枪钻 pistol drill	切割锯 cutting saw	美工刀 utility knife

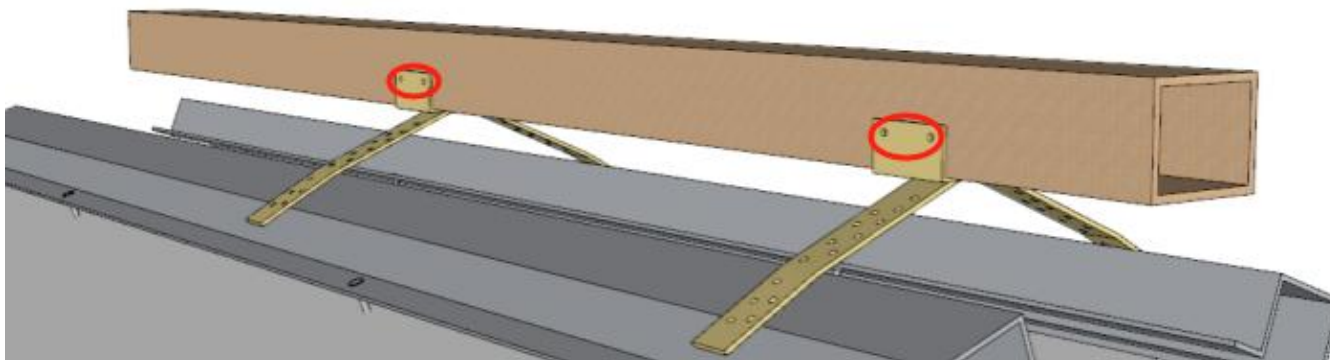
4.6.3.4 安装步骤 INSTALLATION STEPS

1、安装锐固支架，按照下图方式@ $\leq 400\text{mm}$ 布置并用 $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝固定在两侧挂瓦条上

Install the bracket, arrange it @ $\leq 400\text{mm}$ according to the following figure, and fix it on the tile battens on both sides with $\phi 3.8 \times 40\text{mm}$ countersunk drilling screw



2、安装通长龙骨，按照下图布置在锐固支架上， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝通过两侧固定孔固定
Install the long keel, arrange it on the bracket according to the following figure, and fix it with drilling screws through the fixing holes on both sides



3、按照下图方式铺贴通风卷材：

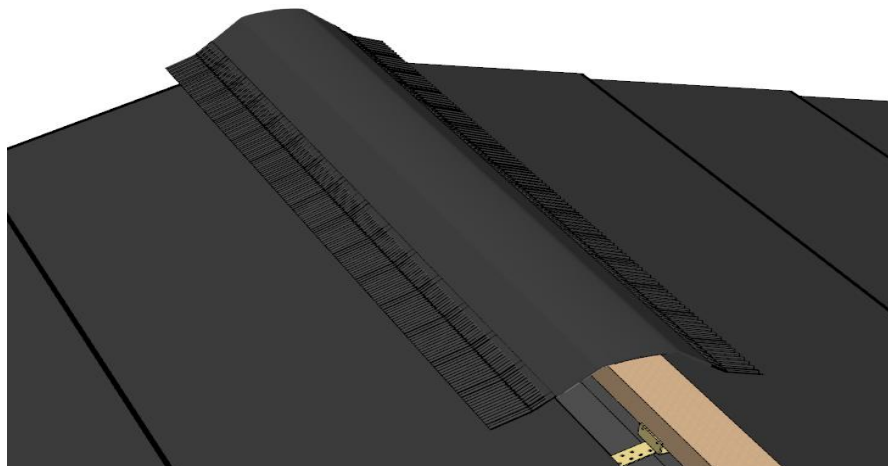
Pave the ventilated membrane as shown below

A、铺贴前需要将两侧主瓦上的灰尘清扫干净

Before paving, the dust on the main tiles on both sides needs to be cleaned

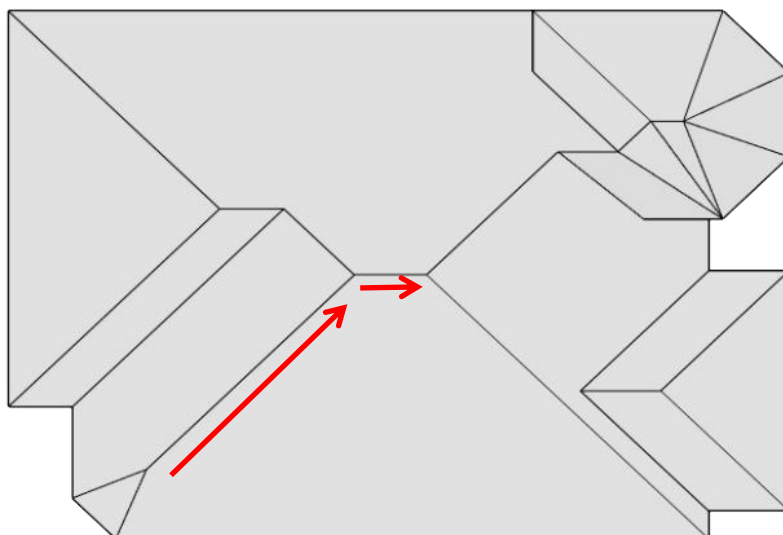
B、撕掉通风卷材两侧的隔离纸

Remove the isolation paper on both sides of the ventilated membrane



4、正脊处由左到右、斜脊处由低到高依此安装脊瓦：

The ridge tiles are installed from left to right and from low to high at the ridge



A、斜脊或正脊端部安装脊瓦封头， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝固定在通长龙骨上

Install ridge tile heads at the ends of inclined or positive ridges, and fix them on the full-length keel with countersunk drilling screw $\phi 3.8 \times 40\text{mm}$



B、连续布置脊瓦到另一端部， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝固定在通长龙骨上

Continuously arrange the ridge tiles to the other end, and fix them on the full-length keel with countersunk drilling screws of $\phi 3.8 \times 40\text{mm}$

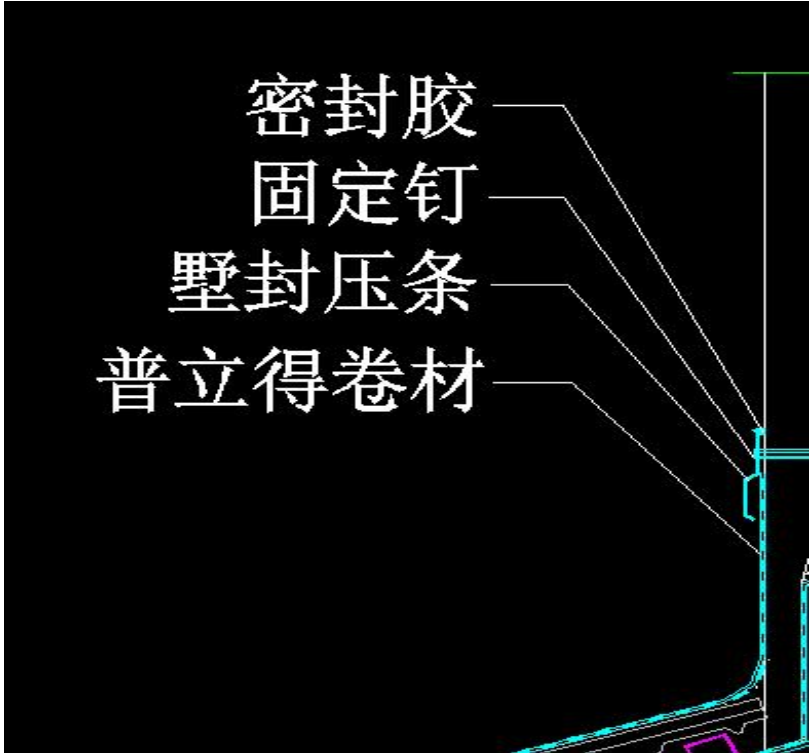


4.6.4 泛水收口 FLASHING

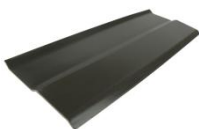
4.6.4.1 注意事项 PRECAUTION

1、在主瓦与山檐的空隙处沿坡度方向固定边瓦龙骨

The edge tile keel is fixed along the slope direction at the gap between the main tiles and the edge tiles

密封胶 sealant	
固定钉 drilling screw	
墅封压条 Edge sealant strip	
普立得卷材 PLD coil materials	

4.6.4.2 工序材料 MATERIALS

			
普立得卷材 PLD coil materials	墅封压条 Edge sealant strip	耐候密封胶 sealant	沉头燕尾丝 drilling screw Φ3.8*40mm

4.6.4.3 使用工具 TOOLS

		
手枪钻 pistol drill	切割锯 cutting saw	胶枪 glue gun

4.6.4.4 安装步骤 INSTALLATION STEPS

A、清理泛水两侧主瓦和墙体的灰尘

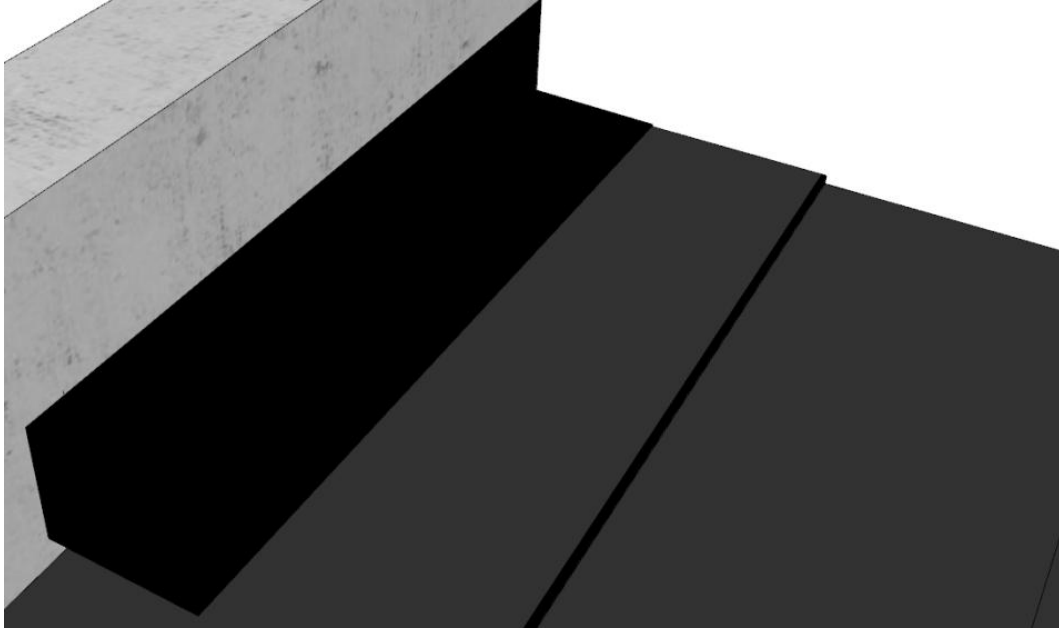
Clean the dust from the main tiles and walls on both sides of the flashing

B、将普立得卷材对折 Fold PLD coil material in half

C、撕掉普立得卷材底部隔离纸 Remove the barrier paper at the bottom of the coil

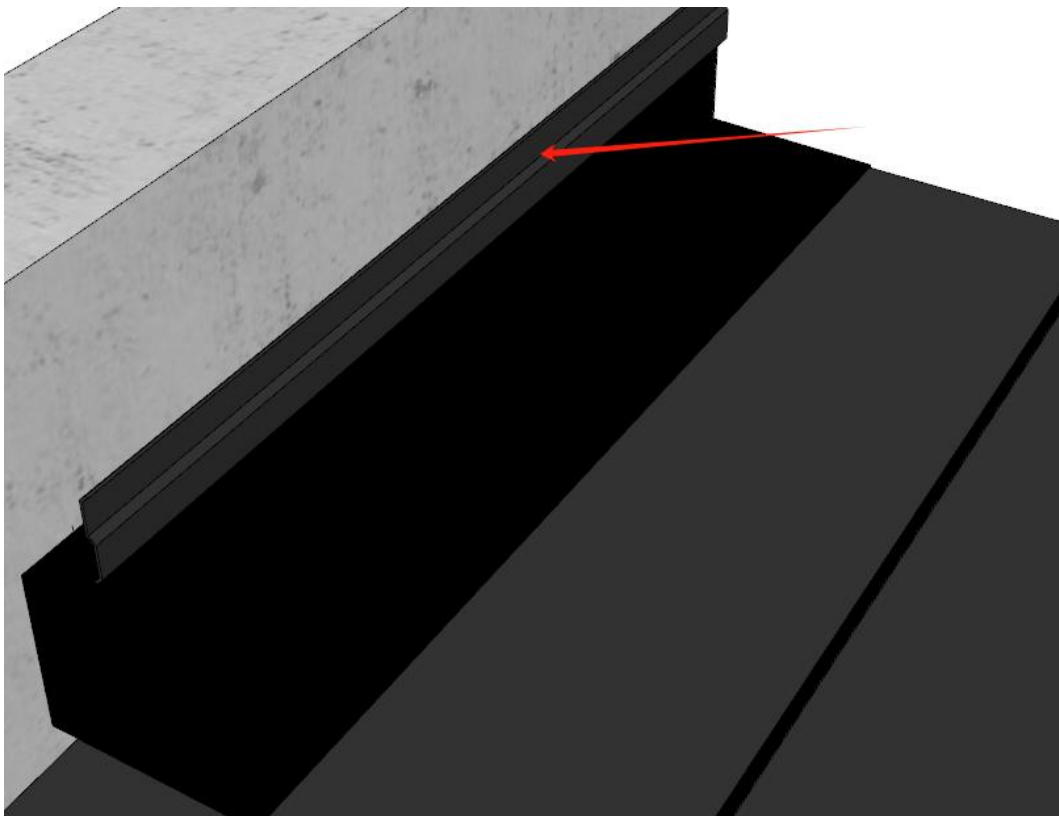
D、将普立得卷材贴于主瓦表面和墙体上

Stick the PLD material onto the surface of the main tile and the wall



E、密封胶条压住普立得卷材上口， $\phi 3.8 \times 40\text{mm}$ 沉头燕尾丝@400mm 固定在墙体上

The Edge sealant strip is pressed against the upper opening of the PLD material, and is fixed on the wall with the $\phi 3.8 \times 40\text{mm}$ countersunk drilling screws



F、耐候密封胶收口竖缝压条

Weather resistant sealant is to use sealing vertical seam with pressure strip.

